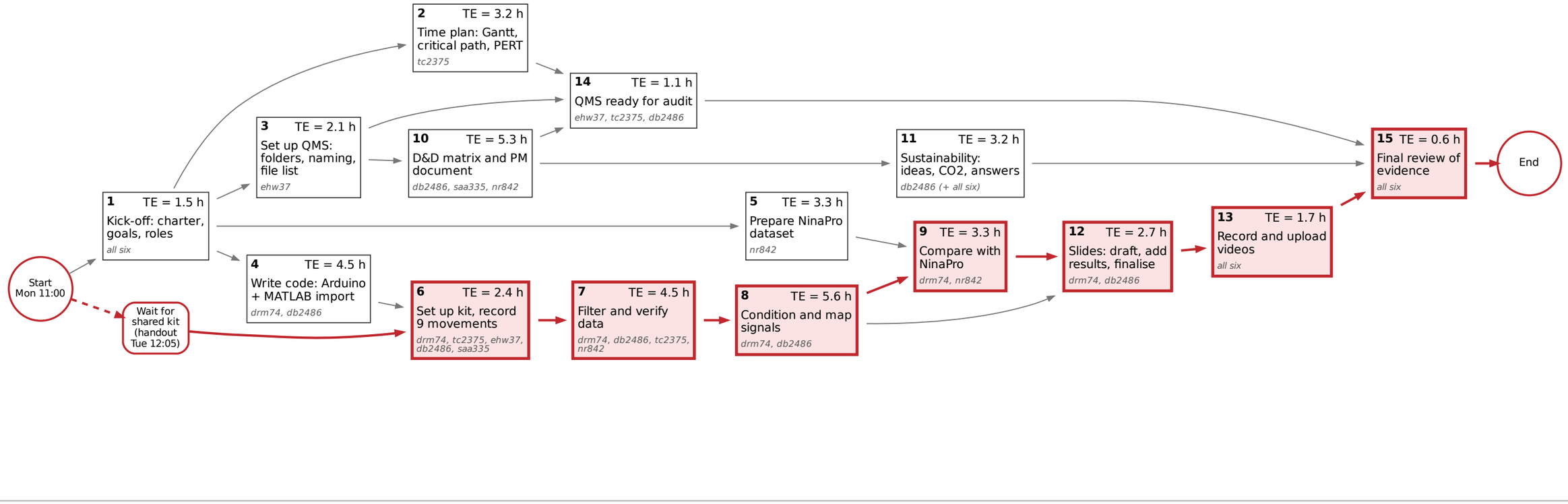
PW1 Group 32 ElectroMyo Gang - PERT chart V3

Activity-on-node network. Each box shows the activity number, its expected time TE, the activity and who did it. Red boxes and arrows = critical path. Times are working hours on the team calendar (08:00-20:00, Thu until 17:00).



Three-point estimates (hours)

ID	Activity	Owners (usernames)	Pred.	O	M	P	TE	σ	Slack	Critical
1	Kick-off: charter, goals, roles	all six	-	1.5	1.5	1.5	1.50	0.00	7.1	
2	Time plan: Gantt, critical path, PERT	tc2375	1	2	3	5	3.17	0.50	27.5	
3	Set up QMS: folders, naming, file list	ehw37	1	1.5	2	3	2.08	0.25	21.1	
4	Write code: Arduino + MATLAB import	drm74, db2486	1	3	4	8	4.50	0.83	7.1	
5	Prepare NinaPro dataset	nr842	1	2	3	6	3.33	0.67	20.8	
6	Set up kit, record 9 movements	drm74, tc2375, ehw37, db2486, saa335	4, kit	1.5	2	5	2.42	0.58	0.0	Yes
7	Filter and verify data	drm74, db2486, tc2375, nr842	6	3	4	8	4.50	0.83	0.0	Yes
8	Condition and map signals	drm74, db2486	7	3.5	5	10	5.58	1.08	0.0	Yes
9	Compare with NinaPro	drm74, nr842	8, 5	2	3	6	3.33	0.67	0.0	Yes
10	D&D matrix and PM document	db2486, saa335, nr842	3	4	5	8	5.33	0.67	21.1	
11	Sustainability: ideas, CO2, answers	db2486 (+ all six)	10	2.5	3	5	3.25	0.42	21.1	
12	Slides: draft, add results, finalise	drm74, db2486	8, 9	2	2.5	4	2.67	0.33	0.0	Yes
13	Record and upload videos	all six	12	1	1.5	3	1.67	0.33	0.0	Yes
14	QMS ready for audit	ehw37, tc2375, db2486	2, 3, 10	0.75	1	2	1.12	0.21	23.2	
15	Final review of evidence	all six	11, 13, 14	0.5	0.5	1	0.58	0.08	0.0	Yes

How the estimates were used

TE = (O + 4M + P) / 6
σ = (P − O) / 6
σ total = √(Σ σ²) along the critical path

Critical path

Start → kit handout (Tue 12:05) → 6 → 7 → 8 → 9 → 12 → 13 → 15 → End
The shared kit is the bottleneck: no recording could start before Tue 12:05.

Expected finish

Kit wait 13.1 h + critical activities 20.8 h = 33.8 h → Thu 08:50
σ total = √(0.58² + 0.83² + 1.08² + 0.67² + 0.33² + 0.33² + 0.08²) = 1.7 h
68% chance of finishing between Wed 19:08 and Thu 10:31 (33.8 ± 1.7 h).

Buffers

Time to the Thu 17:00 deadline = 42 h, so 8.2 h is spare (about 5σ).
This spare time is protected on the Gantt chart as separate buffer tasks:
audit check (Thu 10:00), video re-takes (Thu 14:45) and the project buffer (Thu 16:30).